

	AWS Solution Architect												Course Type	LTP		
Course Code:	CSE3016												Credits	4		
Prerequisite:	AWS Cloud Practitioner															
Course Objectives:																
1. To understand advanced AWS Cloud concepts, services, and architectural principles. 2. To gain expertise in designing and implementing highly available, scalable, and fault-tolerant AWS solutions. 3. To learn best practices for security, compliance, and cost optimization in AWS architecture. 4. To acquire skills in designing and deploying multi-tier applications on AWS infrastructure.																
Course Outcomes (CO):																
At the end of the course, students should be able to																
CO1. To have an in-depth understanding of AWS cloud architecture, service. and design patterns. [KL-3]																
CO2. To design and deploy resilient and scalable multi-tier applications on AWS infrastructure using service like EC2, RDS, S3, and Lambda. [KL-3]																
CO3. To configure and manage advanced AWS networking failure feature, storage solution, database technology and compute resource. [KL-4]																
CO4. To monitor and usage of cloud data and examine resource availability and scale up or down used in autoscaling. [KL-4]																
CO5. To optimize cost and performance by selecting appropriate AWS services and utilizing cost-effective resource management strategies. [KL-3]																
Correlation of COs with POs																
CO \ PO	CKL	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
PKL		3	5	6	5	6	3	3	3	2	2	3	3	3	3	4
CO1	3	3	3	2	3	2	3	3	3	2	2	3	3	3	3	2
CO2	3	3	3	2	3	2	3	3	3	2	2	3	3	3	3	2
CO3	4	3	3	2	3	2	3	3	3	3	3	3	3	3	3	3
CO4	4	3	3	2	3	2	3	3	3	3	3	3	3	3	3	3
CO5	3	3	3	2	3	2	3	3	3	2	2	3	3	3	3	2
CO	Topics to be discussed													Hrs.		
CO1	AWS Cloud Foundations: Introduction to Cloud Computing, Overview of AWS Global Infrastructure, AWS Shared Responsibility Model AWS IAM - AWS Identity and Access Management (IAM), IAM Users, Groups, and Policies, IAM Roles, IAM Best Practices													12		
CO2	Compute – Amazon EC2 (Elastic Compute Cloud), EC2 Instance Types, EC2 Pricing Models, Elastic IP, AWS Elastic Beanstalk, AWS Fargate, Amazon Elastic Container Service (ECS), Amazon Elastic Kubernetes Service (EKS). Storage – AWS Storage Services, Amazon S3 (Simple Storage Service), S3 Storage Classes, S3 Lifecycle Policies, Amazon EBS (Elastic Block Store), Amazon Glacier, Amazon EFS (Elastic File System)													12		
CO3	Networking & Content Delivery – Amazon API Gateway, Amazon CloudFront, Direct Connect, AWS Networking and Content Delivery, Amazon VPC (Virtual Private Cloud), VPC Peering. Security Identification and Compline – AWS Security and Compliance, AWS Shared Responsibility Model, AWS Key Management Service (KMS), AWS WAF (Web Application Firewall), AWS Shield, AWS Security Best Practices													12		

CO4	Database – RDS Multi-AZ Deployments, RDS Read Replicas, Amazon Aurora, Amazon DynamoDB, DynamoDB Streams Management Tools – AWS Management Tools, AWS CloudWatch, CloudWatch Metrics, CloudWatch Alarms, AWS CloudTrail, AWS Config, AWS Systems Manager. Management & Governance – AWS ELB and Auto Scaling, CloudFormation, CloudTrail, CloudWatch.	10
CO5	Analytics – Amazon Athena, AWS Data Exchange, Amazon EMR, AWS Glue, Amazon Kinesis, Quick Sight and Amazon Redshift. AWS Migration and Hybrid Architecture , AWS Database Migration Service (DMS), AWS Server Migration Service (SMS), AWS Snowball, AWS Snowball Edge, AWS Snowmobile	12
Guest Lecture on Contemporary Topics		02
	Total Lecture:	60
Mode of Teaching and Learning: Flipped Classroom, Activity-Based Teaching/Learning, Digital/Computer-based models, wherever possible to augment lecture for practice/tutorial and minimum 2 hours lectures by industry experts on contemporary topics.		
Mode of Evaluation and assessment: The assessment and evaluation components may consist of unannounced open book examinations, quizzes, student's portfolio generation and assessment, and any other innovative assessment practices followed by faculty, in addition to the Continuous Assessment Tests and Term End Examinations.		
Text Books: Piper, Ben, and David Clinton. AWS Certified Solutions Architect Study Guide: Associate SAA-C02 Exam. John Wiley & Sons, 2020.		
Reference books: Baron, Joe, Hisham Baz, Tim Bixler, Biff Gaut, Kevin E. Kelly, Sean Senior, and John Stamper. AWS certified solutions architect official study guide: associate exam. John Wiley & Sons, 2016.		
Indicative list of Experiments: 1. Launch an EC2 instance using the AWS Management Console and deploy a website using Apache middle ware. 2. Deploy a static website in S3 and manage the website using S3 life cycle management. 3. creating DNS records, configuring routing policies, implementing health checks, and monitoring DNS health using Route 53. 4. Create and configure Amazon Elastic File System (EFS) to create scalable and shared file storage for AWS resources and EFS file system, mount it to EC2 instances, and configure access permissions. 5. Create and configure CloudFront distributions to accelerate the delivery of their web content, improve website performance, and enhance user experience. 6. Create AWS Certificate Manager (ACM) for managing SSL/TLS certificates for secure communication in AWS. 7. Create and manage customer master keys (CMKs), encrypt data using KMS, and integrate KMS with AWS services for encryption. 8. Set up CloudWatch monitoring for AWS resources, create custom metrics and alarms, analyze logs, and troubleshoot performance issues using CloudWatch.		
Recommendation by the Board of Studies on		16.04.2024
Approval by Academic council on:		
Prepared by		Dr. C. P. Koushik
Compiled by:		Dr. Sandip Mal